

Bounded-Width Power-Set Compression

Exact finite thresholds, choiceless infinite obstructions,
and a defect extension of Glazer's chain-fiber theorem

A finite/infinite research problem with proof certificates and a formalization route

Prepared for Vladimir Reshetnikov

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Research status. The finite theorems are proved from Dilworth's and Sperner's theorems, with an explicit symmetric-chain certificate. The local-expansion, Dedekind-infinitude, and antichain-transfer theorems are proved in the manuscript; the transfer theorem imports Forster's published finite-to-one strengthening of Cantor's theorem. The full choiceless classification for width bound $r \geq 2$ is posed as an open problem, not claimed solved. Historical priority for the new formulation and intermediate observations has not been established. This manuscript is AI-assisted, unrefereed, and not formally verified. Elliot Glazer is a cited researcher, not an author or endorser.

Repository snapshot. The proposed integration targets the public *ProveIt* repository at commit b71545625, inspected on 5 October 2026. The archive accompanying this manuscript contains the $\text{\LaTeX}$ source, compiled PDF, and an exact standard-library Python verifier.

Abstract

For a poset P and a positive integer r , call a map $c : P \rightarrow Y$ an r -width compression when every fiber has width at most r . The motivating infinite question asks for such a map

$$f : \mathcal{P}(X) \longrightarrow X,$$

where $\mathcal{P}(X)$ is ordered by inclusion. At $r = 1$ this is the chain-fiber problem recently settled in ZF by Elliot Glazer: such a map exists exactly when $1 \leq |X| \leq 3$.

We formulate the bounded-width extension and solve its finite form exactly. For every finite poset P of width w and every list of fiber budgets $r_1, \dots, r_q$, a coloring with i th fiber of width at most r_i exists if and only if $\sum_i r_i \geq w$. Hence, for the Boolean lattice $\mathbb{B}_n = \mathcal{P}([n])$,

$$q_r(n) = \min\{|Y| : \mathbb{B}_n \rightarrow Y \text{ has fibers of width } \leq r\} = \left\lceil \frac{\binom{n}{\lfloor n/2 \rfloor}}{r} \right\rceil.$$

A self-indexed compression $\mathcal{P}([n]) \rightarrow [n]$ therefore exists exactly when $\binom{n}{\lfloor n/2 \rfloor} \leq rn$. Its least admissible defect is

$$\delta(n) = \left\lceil \frac{1}{n} \binom{n}{\lfloor n/2 \rfloor} \right\rceil \sim \sqrt{\frac{2}{\pi}} \frac{2^n}{n^{3/2}},$$

and the largest feasible n for a fixed r is

$$N(r) = \log_2 r + \frac{3}{2} \log_2 \log_2 r + \frac{1}{2} \log_2 \frac{\pi}{2} + O(1).$$

The upper bound is constructive: group the chains in a symmetric-chain decomposition, at most r chains per label. The middle layer is the matching lower-bound certificate.

For the infinite form we prove three choice-free results. First, every finite $S \subseteq X$ obeys the exact expansion inequality

$$|f[\mathcal{P}(S)]| \geq q_r(|S|).$$

Consequently every hypothetical infinite example canonically generates a countably infinite subset by iterating $S \mapsto S \cup f[\mathcal{P}(S)]$; in particular no Dedekind-finite or amorphous set can be a counterexample. Second, an antichain-transfer theorem converts any antichain code $\mathcal{P}(U) \rightarrow \mathcal{P}(X)$ together with a finite-to-one map $X \rightarrow U$ into a finite-to-one map $\mathcal{P}(U) \rightarrow U$, contradicting Forster when U is infinite. Third, if $2 \times X$ injects into X , the complementary-transversal code supplies such an antichain. Thus no well-orderable infinite X works, and under Choice the finite classification is complete. The unresolved ZF frontier for $r \geq 2$ is sharply localized: an infinite counterexample would have to be Dedekind-infinite but non-duplicable, with no suitable finite-to-one core.

Keywords. Boolean lattice; poset width; Dilworth theorem; Sperner theorem; symmetric-chain decomposition; power set; axiom of choice; finite-to-one map; Dedekind-infinite set; formal proof certificate.

[write] **Status and trust boundary, added 5 October 2026, batch 114.** This is an AI-assisted, unrefereed external manuscript (its status box says so; the title page reads “Prepared for Vladimir Reshetnikov / Research manuscript prepared with ChatGPT”), filed in ProveIt as the single-source research report bounded-width-power-set-compression. Provenance, the sources as the write read them, the relation to the repository, the collected non-claims and the reading conventions are in Section 1.6. No statement of the report is formalized in Lean or Rocq; the blueprint of Section 12 is a proposal. *One statement is wrong*

as printed. Definition 9.1 does not require the antichain code Θ to be injective, and with that definition Lemma 9.2, Theorem 9.4 and the abstract's sentence that the transfer theorem "converts any antichain code" into a finite-to-one map are false: Remark 9.10 gives the counterexample $X = \{0, 1, 2\}$, $r = 1$, $U = \mathbb{N}$, $\Theta \equiv \emptyset$ and the repair (require Θ injective). The only code the manuscript constructs, that of Lemma 9.5, is injective, so every later result and the complete ZFC classification stand. *What rests on what.* Dilworth's and Sperner's theorems are quoted; Glazer's $r = 1$ theorem (Theorem 7.1) is imported from a MathOverflow answer, which the write read; Forster's theorem (Theorem 9.3) is imported, and the write read only its abstract (note after the theorem). Everything else is proved in the text; the intake and the write reread every proof and found no other error. The program code/verification.py checks finite instances only. The ZF classification for $r \geq 2$ (Conjecture 2.5) is open; it is the first of the source's sixteen research items, all of which are stated as open questions in Section 13.1, which also proves the bipartite-graph assertion of item 6 (Proposition 13.1).

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1 The problem and why the finite/infinite bridge matters

1.1 The chain-fiber starting point

In December 2025 Alexander Pruss asked whether ZF alone rules out, for an infinite set X , a map $f : \mathcal{P}(X) \rightarrow X$ that separates every incomparable pair: $f(A) \neq f(B)$ whenever neither $A \subseteq B$ nor $B \subseteq A$. Equivalently, each fiber $f^{-1}(x)$ must be a chain under inclusion. Elliot Glazer answered with a sharp theorem: in ZF, the map exists exactly for sets of cardinality 1, 2, or 3 [9]. Emil Jeřábek supplied a complementary cardinal-arithmetic argument on the same thread.

That result has an unusually clean finite/infinite profile. The finite cutoff is not merely an artifact of cardinality: it is the width of a Boolean lattice. The infinite proof, by contrast, must survive the absence of a well-order and of the finite choices that would normally split a bounded-width poset into chains. This suggests varying the amount of incomparability allowed inside one label.

1.2 The bounded-width extension

Fix a positive integer r . We permit each label to identify several mutually incompatible subsets, but cap the size of an antichain inside one fiber by r .

Problem 1.1 (bounded-width power-set compression). Classify the pairs (X, r) for which there is a function

$$f : \mathcal{P}(X) \longrightarrow X$$

such that every fiber $f^{-1}(x)$ has inclusion-width at most r . Determine the exact finite threshold, the strongest theorem provable in ZF, and the additional strength supplied by choice principles.

The parameter r is a *contradiction budget*: a label may be attached to at most r pairwise incompatible descriptions. The case $r = 1$ is Glazer’s theorem. For $r > 1$ the problem retains a fully explicit finite extremal answer but opens a genuinely choiceless infinite frontier.

1.3 Why this should interest Glazer

The problem directly extends a theorem proved by Glazer, while its infinite part lies at the intersection of his documented interests in choiceless set theory and hard, independently checkable mathematical problems. His work with Bokai Yao separates principles that coincide under Choice but diverge in ZF with urelements [5]; the present problem similarly has a complete ZFC classification and a narrower unresolved ZF boundary. His work on FrontierMath emphasizes original expert-level problems with objective verification surfaces [4]; here the finite theorem has compact upper and lower certificates, while the infinite theorem has a precise imported dependency and a transparent open remainder.

1.4 Relation to ProveIt

The current ProveIt repository describes itself as machine-checked mathematics in Lean and Rocq/Coq, with executable certificates separated from untrusted exploratory computations [8]. This problem naturally spans three of its active surfaces:

1. finite posets and Boolean-lattice certificates in combinatorics;

2. first-order ZF and cardinal arithmetic in set theory;
3. a small proof-producing checker whose trusted boundary is simpler than a general-purpose optimizer.

The finite core is suitable for immediate formalization. The infinite side suggests a staged project: formalize the local expansion and antichain code first, then isolate Forster’s theorem as the only substantial imported set-theoretic component.

[write] Scope of the repository claim, 5 October 2026. The first sentence is correct: the root README.md of the pinned snapshot b71545625 (unchanged at the write’s base commit bf70d7db4) describes ProveIt as machine-checked mathematics in Lean 4 and Rocq/Coq whose generated data and exploratory computations lie outside the trusted theorem boundary unless a proved checker connects them. The list of “active surfaces” overstates what the repository has. *Item 1 does not exist:* at the pin and at the base commit no *.lean or *.v file contains Dilworth’s or Sperner’s theorem, the LYM inequality, a symmetric-chain decomposition, a Peck poset, or the width of a finite poset (git grep of all Lean and Rocq sources: no hit for Dilworth, Sperner, symmetric chain, LYM, Lubell or Peck). “Boolean lattice” occurs only in doc comments of Analysis/FabiusFunction/Lean/FabiusFunction/ThueMorseBooleanMobius.lean (signed sums over intervals of the lattice of submasks), and antichains only in well-quasi-order arguments (Dickson’s lemma in the surreal and transseries libraries) and in the chain condition of forcing posets (SetTheory/Cardinals/Cardinals/Above.lean). The pin’s Combinatorics/ holds derivative expansions, expression enumeration, polyominoes, Ramsey theory and squared squares. *Item 2 exists only in part:* first-order ZF is formalized in SetTheory/ZF (Lean and Rocq), and cardinal arithmetic only as the ZFC large-cardinal library SetTheory/Cardinals/Cardinals; nothing in the repository treats choiceless cardinal arithmetic, Dedekind-finite or amorphous sets, finite-to-one maps or Forster’s theorem. So this report would *open* those surfaces rather than continue them; Sperner’s theorem and the LYM inequality are available to a Lean development only through mathlib (note in Section 12).

1.5 Contribution boundary

The exact finite formula is a short synthesis of classical results of Dilworth and Sperner, enhanced here by a nonuniform budget theorem, an explicit certificate format, threshold inversion, and its connection to the choiceless problem. The local-expansion, definable Dedekind-infinitude, and general antichain-transfer statements are proved below. A targeted search of the inspected literature did not locate this precise bounded-width formulation, but absence from a search is not a priority proof. No claim is made that the unresolved $r \geq 2$ question is known to be open in every source not inspected.

Main outcome. The finite problem and the entire ZFC classification are solved exactly. In ZF, every possible infinite counterexample for $r \geq 2$ is forced into a narrow cardinal-arithmetic gap: it is Dedekind-infinite, it is not binary-duplicable, and it has no finite-to-one core carrying two disjoint copies of itself.

1.6 Provenance, sources and reading conventions

[write] Added 5 October 2026, batch 114. *Provenance.* The report is built from one manuscript, the archive Bounded_Width_Power_Set_Compression.zip (six files at the archive root, no wrapper directory), which reached ProveIt’s docs/incoming in the arrival commit 2399df2bd (5 October 2026, “New research reports”), outside the session bundle of the intake, and was placed as batch 114 in 99053b5d1. The text is the delivered article.tex (1,288 lines), shipped under the same name; the delivered verification.py and verification_report.txt

are shipped unchanged in code/ and data/. The delivered 21-page PDF and SHA256SUMS.txt (four entries, all verified at placement) are not shipped, so the title page’s “the archive accompanying this manuscript contains the $\text{\LaTeX}$ source, compiled PDF, and ... verifier” describes the delivery, not this directory; they and the delivery README.md (staged at placement and replaced by the report README) are retrievable from the arrival commit. The package pins ProveIt revision b71545625fedc631edc2ead542abbfba72eb6063 (bibliography entry [8]); that commit is the batch-98B write commit of this repository and an ancestor of the commit bf70d7db4 this write is based on. The package contains no e-mail address, no “prepared for private review” line and no personal data; “Prepared for Vladimir Reshetnikov” and “prepared with ChatGPT” record authorship. Appendix C names a tool of the generating assistant’s environment (note there). Single source: no merge choice was made. The write prefixed the 51 delivered labels with bwc: and updated every reference; it added the notes marked [write], this subsection, labels on four delivered headings or remarks and on the sixteen research items, Remark 9.10 (the last numbered statement of Section 9), and Section 13.1 with Proposition 13.1. No statement, proof or number of the source was changed; the added displays are unnumbered, so every delivered statement and equation keeps its number.

The sources, as the write read them (5 October 2026).

- The MathOverflow thread [9]: read in full through the Stack Exchange API (question and all three answers; content licensed CC BY-SA 4.0). It confirms Theorem 7.1 in the printed form and contains the coding behind Lemma 9.5 (note after Theorem 7.1).
- Forster [3]: abstract and bibliographic data read on the publisher’s page; the body was not read (note after Theorem 9.3).
- Hu and Shen [7]: the arXiv abstract read; it states ZF nonexistence theorems for boundedly finite-to-one functions (from all permutations of an infinite set to those moving at most n points, and from finitary partitions to finite subsets). The two sentences Section 7 attributes to the paper (a boundedly finite-to-one map need not yield an injection; models where familiar implications fail) were not checked against its body.
- Not read: Dilworth [2], Sperner [11], de Bruijn–van Ebbenhorst Tengbergen–Kruyswijk [1], Greene–Kleitman [6], Shen [10] (title and data only), Glazer–Yao [5] and FrontierMath [4] (motivation only). The standard facts taken from the first four (Dilworth’s finite chain partition, Sperner’s width theorem, the existence of a symmetric-chain decomposition) are used as the source states them; the recursive decomposition of Section 4 is proved in the text and checked by the program for $n \leq 14$.

What was checked at intake and at the write. The batch-114 dossier read the manuscript in full and found correct the budget theorem, the symmetric-chain recursion, the ratios $2m/(m+1)$ and $(2m+1)/(m+1)$, the inversion, the local expansion, the hull recursion and its choice-free ordering, Lemma 9.5, Corollary 9.8 and Theorem 10.1, and found the injectivity gap of Remark 9.10; independent programs confirmed $\delta(1), \dots, \delta(16)$, the $N(r)$ table, and the witness for $\text{SC}_1(\{0, 1, 2\})$. The delivered verifier, run on a copy, printed its recorded output. The write reread every proof (only the gap of Remark 9.10 found), reran the verifier on a copy (under a second, output identical up to line endings), and checked with its own program the counterexample in full, both tables of Appendix A (all 24 rows of Table 1, all 18 entries of Table 2, including $N(26) = 10$, which the verifier does not print), the exact ratio identities for $1 \leq m \leq 200$, and Example 4.2 against the recursive decomposition.

Relation to the repository.

- *Formal status.* No statement of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status. The blueprint of Section 12 is a proposal.
- *Stale claims.* The claim of Section 1.4 that the problem spans existing “active surfaces” of finite posets and Boolean-lattice certificates is overstated; the note there scopes it. The other two statements about the repository are correct.
- *ordinals-and-order-types/freiling-symmetry-blacklists* (same category and batch; *see also*): another exact finite extremal theory whose infinite form is located between choice principles over ZF (there, the Boolean prime ideal theorem implies the partition principle, which implies choice for families of

m -element sets, $2 \leq m \leq k + 1$). It too asks for the weakest standard choice principle that suffices (here item 3 of Section 13). Neither source cites the other; no theorem is shared.

- ordinals-and-order-types/naming-elementary-embeddings cites the same Glazer–Yao paper [5], which is motivation only here. Batch 114 also filed noisy-parity-cubes, robust-neutral-choice and random-bits-arithmetic-cuts in the same category; they share only the Glazer motivation, and no theorem.

[write] What is not claimed, collected from the source. The exact finite formula is “a short synthesis of classical results of Dilworth and Sperner”. “Absence from a search is not a priority proof”, and “no claim is made that the unresolved $r \geq 2$ question is known to be open in every source not inspected”; historical priority “has not been established”. Glazer’s theorem is imported from a MathOverflow answer and “not reproved in full”; Forster’s theorem is imported. The full ZF classification for $r \geq 2$ “is posed as an open problem, not claimed solved”. The verifier’s role “is diagnostic, not foundational”. The manuscript is “AI-assisted, unrefereed, and not formally verified”, and “Elliot Glazer is a cited researcher, not an author or endorser”. The research questions and the formalization blueprint are proposals. The write adds one more: the antichain-transfer theorem holds only for injective codes (Remark 9.10).

[write] Reading conventions. “Proved here” in the ledger of Appendix B means proved in this text, not first proved here; “Imported” means quoted without proof. $[n] = \{1, \dots, n\}$ and $2 = \{0, 1\}$; $X \times 2$ and $2 \times X$ are interchangeable (a swap is a bijection). The width of an infinite poset is used only through “ $\text{wd}(P) \leq r$ ”, which Definition 2.1 reads as “no antichain of size $r + 1$ ”. A *label* is an element of the codomain of a compression and a *fiber* is a preimage of one label. A map is *finite-to-one* if every fiber is finite, and *at most r -to-one* if every fiber has at most r elements. The source reuses or leaves undefined the following symbols; none was renamed.

Symbol	Meaning here (where)	Not to be read as
$U \preceq X$	there is an injection $U \rightarrow X$ (Theorem 9.6)	
$X \preceq_{\text{fto}} U$	there is a finite-to-one map $X \rightarrow U$ (Theorems 9.6, 10.1; used, never defined in the text; the macro <code>\bf to</code> is defined and never used)	a map $U \rightarrow X$; an injection
$q, q_r(n)$	the number of bins in Theorem 3.1 and a claimed number of labels in Section 4; the least number of labels for $\mathbb{B}_n$ (Definition 2.3); also the alphabet size of item 11	each other
$\chi_r(P)$	the least number of labels of an r -width compression of P (Definition 2.2)	the chromatic number of a graph
W_n, L_n	$\binom{n}{\lfloor n/2 \rfloor}$; the middle layer of $\mathbb{B}_n$	$L = \log_2 r$
L	$\log_2 r$ (Theorem 6.2)	the middle layer L_n
$\delta(n), N(r)$	the least admissible width budget for $\mathcal{P}([n]) \rightarrow [n]$; the largest feasible n for budget r	inverse functions; only $\delta(n) \leq r \iff n \leq N(r)$ holds ($n \geq 1$, by Proposition 5.3)
r, r_i	the uniform width budget; the budget of bin i	
H_f, S_k, μ_k, x_k	the finite hull, its iterates, least preimages, the new points (Theorem 8.3)	
Θ, e, e_i	an antichain code (Definition 9.1, to be read as injective, Remark 9.10); the injection $U \times 2 \rightarrow X$ and its two channels (Lemma 9.5)	
p, h	the finite-to-one map $X \rightarrow U$; $h = f \circ \Theta$ (Theorem 9.4)	
$C, C^\pm, C_i$	chains of a symmetric-chain decomposition	
c, c_n	a compression or colouring; the points of a definable countable subset (item 6)	each other
$\text{SC}_r(X)$	Definition 2.4	

2 Definitions and the classification targets

Definition 2.1 (width). For a poset P , its width $\text{wd}(P)$ is the supremum of the cardinalities of its antichains. When $r \in \mathbb{N}_{>0}$, the assertion $\text{wd}(P) \leq r$ means that P has no antichain of size $r + 1$.

Definition 2.2 (width compression). Let P be a poset, let Y be a set, and let $r \geq 1$. An r -width compression is a function $c : P \rightarrow Y$ such that

$$\text{wd}(c^{-1}(y)) \leq r \quad (y \in Y).$$

For a finite poset define

$$\chi_r(P) = \min\{|Y| : \text{an } r\text{-width compression } P \rightarrow Y \text{ exists}\}.$$

Definition 2.3 (Boolean and self-indexed parameters). Write $\mathbb{B}_n = (\mathcal{P}([n]), \subseteq)$ and

$$W_n = \text{wd}(\mathbb{B}_n) = \binom{n}{\lfloor n/2 \rfloor}.$$

Set

$$q_r(n) = \chi_r(\mathbb{B}_n), \quad \delta(n) = \min\{r : \mathcal{P}([n]) \rightarrow [n] \text{ is an } r\text{-width compression}\},$$

for $n \geq 1$, and

$$N(r) = \max\{n \geq 1 : \delta(n) \leq r\}.$$

For arbitrary sets it is useful to name the existence statement.

Definition 2.4 (self-compression statement). For $r \geq 1$, let $\text{SC}_r(X)$ denote:

$$\exists f : \mathcal{P}(X) \rightarrow X \quad \forall x \in X \quad \text{wd}(f^{-1}(x)) \leq r.$$

The strongest natural classification suggested by the finite answer is the following.

Conjecture 2.5 (bounded-width Glazer classification). For every fixed $r \geq 1$, ZF proves that $\text{SC}_r(X)$ holds if and only if X is finite, nonempty, and

$$\binom{|X|}{\lfloor |X|/2 \rfloor} \leq r^{|X|}.$$

For $r = 1$ this is Glazer's theorem. We prove the finite direction in ZF, the full statement in ZFC, and several choice-free restrictions on a hypothetical counterexample for $r \geq 2$.

3 A universal finite theorem with nonuniform budgets

The first result is not specific to Boolean lattices. It identifies width as the complete resource controlling finite compression.

Theorem 3.1 (budgeted finite-poset theorem). *Let P be a finite poset of width w , and let $r_1, \dots, r_q$ be nonnegative integers. There is a map $c : P \rightarrow [q]$ satisfying*

$$\text{wd}(c^{-1}(i)) \leq r_i \quad (1 \leq i \leq q)$$

if and only if

$$\sum_{i=1}^q r_i \geq w.$$

Proof. Let A be an antichain of size w . Each fiber contains at most r_i points of A , so

$$w = |A| = \sum_{i=1}^q |A \cap c^{-1}(i)| \leq \sum_{i=1}^q r_i.$$

This proves necessity.

For sufficiency, Dilworth's decomposition theorem partitions P into exactly w chains [2]. Distribute these chains among bins $1, \dots, q$, placing at most r_i chains in bin i ; the capacity inequality makes this possible. Color every point by the bin containing its chain. An antichain meets each chain at most once, so the union of at most r_i chains has width at most r_i . $\square$

Corollary 3.2 (uniform compression number). *For every finite poset P and every $r \geq 1$,*

$$\chi_r(P) = \left\lceil \frac{\text{wd}(P)}{r} \right\rceil.$$

Proof. Apply [theorem 3.1](#) with all capacities equal to r . $\square$

Remark 3.3 (partition versus cover). The same criterion holds if one begins with a cover by width-bounded subposets. Choose one covering set for each point and delete the point from all other sets; taking subposets cannot increase width. Thus the function and cover formulations are equivalent in the finite theorem.

Remark 3.4 (why the proof does not settle the infinite problem). Two finite ingredients become delicate without Choice. A finite-width infinite poset need not come equipped with a selectable decomposition into finitely many chains, and even if each individual fiber can be decomposed, choosing coherent decompositions for an X -indexed family is itself a choice problem. More importantly, replacing X by $X \times r$ can change cardinality in ZF.

4 Boolean lattices: exact formula and explicit certificates

Sperner's theorem gives $\text{wd}(\mathbb{B}_n) = W_n$ [11]. Combining it with [corollary 3.2](#) yields the exact finite answer.

Theorem 4.1 (exact Boolean-lattice compression). *For all $n \geq 0$ and $r \geq 1$,*

$$q_r(n) = \left\lceil \frac{W_n}{r} \right\rceil = \left\lceil \frac{1}{r} \binom{n}{\lfloor n/2 \rfloor} \right\rceil.$$

The proof has a particularly transparent pair of certificates.

4.1 Lower certificate: the middle layer

The family

$$L_n = \{A \subseteq [n] : |A| = \lfloor n/2 \rfloor\}$$

is an antichain of size W_n . A color can contain at most r members of L_n , so at least $\lceil W_n/r \rceil$ colors are required. The certificate is simply the list of all middle-layer subsets.

4.2 Upper certificate: grouped symmetric chains

The Boolean lattice has a symmetric-chain decomposition: a partition into saturated chains whose bottom and top ranks sum to n [1, 6]. Every such decomposition has exactly W_n chains, because each chain meets rank $\lfloor n/2 \rfloor$ exactly once. Group at most r chains under each label. Every fiber is then a union of at most r chains and consequently has width at most r .

A recursive decomposition can be constructed without optimization. Begin with the sole chain $\emptyset$ in $\mathbb{B}_0$. Suppose

$$C : A_k \subset A_{k+1} \subset \cdots \subset A_{n-1-k}$$

is a symmetric chain in $\mathbb{B}_{n-1}$. Replace it in $\mathbb{B}_n$ by

$$\begin{aligned} C^+ &: A_k \subset \cdots \subset A_{n-1-k} \subset A_{n-1-k} \cup \{n\}, \\ C^- &: A_k \cup \{n\} \subset \cdots \subset A_{n-2-k} \cup \{n\}, \end{aligned}$$

where the second chain is omitted when empty. The two new chains are symmetric, disjoint, and together contain both copies of every old member. Induction gives a full decomposition.

Example 4.2 ($\mathbb{B}_4$). The recursive construction gives six chains:

$$\begin{array}{ll} C_1 : \emptyset < 1 < 12 < 123 < 1234, & C_2 : 4 < 14 < 124, \\ C_3 : 3 < 13 < 134, & C_4 : 34, \\ C_5 : 2 < 23 < 234, & C_6 : 24. \end{array}$$

Here a string denotes the corresponding subset. For $r = 2$, the groups (C_1, C_2) , (C_3, C_4) , and (C_5, C_6) give a three-label compression. The six two-subsets form the matching lower certificate, so three is optimal.

4.3 A proof-certificate interface

An upper certificate consists of a list of chains and a group number for each chain. A checker verifies:

1. every element of $\mathbb{B}_n$ occurs exactly once;
2. consecutive members of each chain differ by adding one element;
3. each group contains at most r chains.

The checker never computes a width. The lower certificate is the middle layer and the integer inequality $|L_n| > r(q - 1)$. Thus optimality can be kernel-checked by elementary finite-set operations.

5 Self-indexed finite compression

A map $\mathcal{P}([n]) \rightarrow [n]$ may leave labels unused. Therefore it exists exactly when the minimum number of labels is at most n .

Theorem 5.1 (complete finite classification). *Let $n \geq 1$ and $r \geq 1$. The following are equivalent:*

1. *there is an r -width compression $\mathcal{P}([n]) \rightarrow [n]$;*
2. $q_r(n) \leq n$;
3. $W_n \leq rn$;
4. $r \geq \delta(n)$, *where*

$$\delta(n) = \left\lceil \frac{W_n}{n} \right\rceil.$$

Proof. Only the equivalence between (1) and (2) needs comment. A map to $[n]$ uses at most n nonempty fibers, so (1) implies $q_r(n) \leq n$. Conversely, an optimal compression with $q_r(n) \leq n$ labels can inject its label set into $[n]$; the remaining labels have empty fibers. $\square$

Corollary 5.2 (the first defects). *The least width budgets are*

$$\delta(1), \delta(2), \dots = 1, 1, 1, 2, 2, 4, 5, 9, 14, 26, 42, 77, 132, 246, 429, 805, \dots$$

In particular:

1. $r = 1$ permits exactly $n = 1, 2, 3$;
2. $r = 2$ permits exactly $1 \leq n \leq 5$;
3. $r = 3$ still permits only $1 \leq n \leq 5$;
4. $r = 4$ permits exactly $1 \leq n \leq 6$.

The finite $r = 1$ line coincides with Glazer's full ZF theorem, which is a useful consistency check on the formulation.

5.1 Monotonicity

The threshold is genuinely one-sided; no later value of n becomes feasible after failure.

Proposition 5.3 (monotonic defect). *The sequence W_n/n is nondecreasing for $n \geq 1$. Hence $\delta(n)$ is nondecreasing and*

$$N(r) = \max\{n : W_n \leq rn\}$$

is well-defined.

Proof. Put $a_n = W_n/n$. For $m \geq 1$,

$$\frac{a_{2m+1}}{a_{2m}} = \frac{2m}{m+1} \geq 1, \quad \frac{a_{2m+2}}{a_{2m+1}} = \frac{2m+1}{m+1} > 1.$$

Also $a_1 = a_2 = 1$. Taking ceilings preserves monotonicity. Finally, $W_n \geq 2^n/(n+1)$ because the largest binomial coefficient is at least the average of the $n+1$ coefficients. Thus $W_n/n \rightarrow \infty$, so the defining set for $N(r)$ is finite. $\square$

Remark 5.4 (plateaus). Ceiling creates a few initial plateaus: $\delta(1) = \delta(2) = \delta(3) = 1$ and $\delta(4) = \delta(5) = 2$. Thereafter the values grow quickly. The real ratio W_n/n itself is strictly increasing except for the first two parity steps where the displayed ratios equal one.

[write] Reading note, 5 October 2026. Precisely: $a_1 = a_2 = a_3 = 1$, and $a_n = W_n/n$ is strictly increasing from $n = 3$ on. The two steps with ratio one are a_2/a_1 (from $a_1 = a_2 = 1$, not one of the displayed ratios) and $a_3/a_2 = 2m/(m+1)$ at $m = 1$; for $m \geq 2$ both displayed ratios exceed one.

6 Asymptotic defect and inversion

The exact formula makes the growth transparent. Stirling's expansion, uniform across the two parities, gives

$$W_n = 2^n \sqrt{\frac{2}{\pi n}} \left(1 + O(n^{-1})\right). \quad (6.1)$$

Theorem 6.1 (defect asymptotic). *As $n \rightarrow \infty$,*

$$\delta(n) = \sqrt{\frac{2}{\pi}} \frac{2^n}{n^{3/2}} \left(1 + O(n^{-1})\right).$$

Proof. Divide [equation \(6.1\)](#) by n . The ratio tends to infinity, so taking the ceiling changes it by a relative $o(1)$ amount, absorbed by the stated error. $\square$

Thus a fixed multiplicative allowance in fiber width buys only an additive amount in the universe size. More precisely:

Theorem 6.2 (threshold inversion). *Let $L = \log_2 r$. As $r \rightarrow \infty$,*

$$N(r) = L + \frac{3}{2} \log_2 L + \frac{1}{2} \log_2 \frac{\pi}{2} + O(1).$$

Proof. At the smooth transition $W_n/n \asymp r$, [equation \(6.1\)](#) gives

$$n - \frac{3}{2} \log_2 n + \frac{1}{2} \log_2 \frac{2}{\pi} = L + O(n^{-1}).$$

The equation first implies $n = L + O(\log L)$. Substituting this estimate inside $\log_2 n$ yields

$$n = L + \frac{3}{2} \log_2 L + \frac{1}{2} \log_2 \frac{\pi}{2} + O\left(\frac{\log L}{L}\right)$$

for the real transition. The integer maximum $N(r)$ differs from that transition by at most a bounded rounding and parity term, which gives the stated $O(1)$ remainder. $\square$

[write] The remainder, 5 October 2026. The $O(1)$ is not made explicit: the last step of the proof is one sentence, and no bound on the constant is given. The verifier’s diagnostic for $r = 2^k$, $4 \leq k \leq 24$ (Section 11), gives differences $N(r) - (L + \frac{3}{2} \log_2 L + \frac{1}{2} \log_2 \frac{\pi}{2})$ between -0.2032 (at $k = 24$) and 1.6743 (at $k = 4$); the write’s extension to $k \leq 60$ gives values between -0.5636 (at $k = 45$) and the same maximum. These numbers prove nothing; an explicit remainder is item F17 of Section 13.1.

Remark 6.3 (interpretation). The factor $n^{-3/2}$ has two sources: $n^{-1/2}$ from concentration of the binomial layers and another n^{-1} because the available label set has only n elements. Inverting $2^n/n^{3/2}$ creates the secondary $\frac{3}{2} \log_2 \log_2 r$ term.

7 The infinite problem in ZF

7.1 The solved chain case

The following theorem is the exact $r = 1$ endpoint motivating the present extension.

Theorem 7.1 (Glazer, 2025). *ZF proves, for every set X , that the following are equivalent:*

1. $1 \leq |X| \leq 3$;
2. *there is $f : \mathcal{P}(X) \rightarrow X$ whose fibers are inclusion-chains;*
3. *$\mathcal{P}(X)$ is the union of an X -indexed family of inclusion-chains.*

The published form here is a MathOverflow answer rather than a journal paper [9]. The proof is choice-free and uses a diagonal construction after extracting enough coding power from the two-element layer. It is stronger than merely observing that the finite width W_n exceeds n for $n \geq 4$.

[write] The MathOverflow thread, read 5 October 2026. Question 504570 was asked by Alexander Pruss on 2 December 2025. Elliot Glazer’s accepted answer of the same day states that ZF proves, for all sets X , the equivalence of the three conditions of Theorem 7.1, in that form, and proves the hard implication with an injection from $X \sqcup 2$ into the two-element subsets and a diagonal set. Emil Jeřábek’s answer of the same

day completes the argument of the question, which Pruss had given under the hypothesis $2 \cdot \lambda \leq |X| \leq \lambda^2$ for some λ ; Pruss added a second elementary proof on 4 December 2025. *Credit.* The code of Lemma 9.5 is already in that thread for $r = 1$: the question maps $A \subseteq U$ to $f(A \cup (V \setminus g[A]))$ for an injection g from U into a disjoint $V \subseteq X$, and Pruss's answer maps A to $f(h_0[A] \cup (X_1 \setminus h_1[A]))$, noting that the coded sets are incomparable for $A \neq B$; with chain fibers this gives an injection $\mathcal{P}(U) \rightarrow X$ and Cantor's theorem (or, in the question, a cardinal inequality) finishes. The manuscript credits Jeřábek's argument but not this code. What Section 9 adds is the passage from width 1 to width r : the coded map is then only at most r -to-one, and Forster's theorem replaces Cantor's.

7.2 Why $r \geq 2$ is structurally different

For one chain, each rank contains at most one member, and the two-element layer supplies a single-valued coding object. A width- r fiber can contain r members of the same rank. Turning those finite sets into r separately named chains requires choosing an ordering or coloring inside many finite fibers. Such finite choices cannot be silently made in ZF.

This is exactly the setting in which boundedly finite-to-one maps become subtle. Hu and Shen emphasize that, without Choice, a boundedly finite-to-one map need not yield an injection; they prove several generalized Cantor nonexistence theorems while recording models where familiar implications fail [7]. Shen's work on finitary partitions likewise shows that a natural Cantor analogue can consistently fail when the power-set functor is replaced by another large combinatorial construction [10].

The next two sections isolate what can nevertheless be proved directly from the order structure of $\mathcal{P}(X)$.

8 Finite expansion inside every infinite candidate

The finite theorem applies to every finite subcube, even though its labels may land anywhere in X .

Theorem 8.1 (exact local expansion). *Assume $\text{SC}_r(X)$, witnessed by $f : \mathcal{P}(X) \rightarrow X$. For every finite $S \subseteq X$ with $|S| = n$,*

$$|f[\mathcal{P}(S)]| \geq q_r(n) = \left\lceil \frac{W_n}{r} \right\rceil.$$

Consequently,

$$|f[\mathcal{P}(S)] \setminus S| \geq \max \left(0, \left\lceil \frac{W_n}{r} \right\rceil - n \right).$$

Proof. Restrict f to the finite Boolean lattice $\mathcal{P}(S)$. For every label x , the restricted fiber $f^{-1}(x) \cap \mathcal{P}(S)$ is a subposet of the global fiber, hence has width at most r . By theorem 4.1, the restricted map uses at least $q_r(n)$ distinct labels. At most n of those labels lie in S , giving the second inequality. $\square$

Corollary 8.2 (no large finite closed set). *If $n > N(r)$, no n -element set $S \subseteq X$ can satisfy $f[\mathcal{P}(S)] \subseteq S$.*

Proof. For $n > N(r)$, $q_r(n) > n$ by definition. The image cannot both have more than n elements and be contained in S . $\square$

Define the finite-hull operator

$$H_f(S) = S \cup f[\mathcal{P}(S)].$$

Then [theorem 8.1](#) gives the exact recurrence

$$|H_f(S)| \geq q_r(|S|) \quad \text{whenever } |S| > N(r). \quad (8.1)$$

Since $W_n \geq 2^n / (n + 1)$, this eventually dominates $2^n / (r(n + 1))$; iterating the hull therefore grows faster than an ordinary exponential recurrence.

Theorem 8.3 (definable Dedekind-infinitude). *Fix $r \geq 1$. In ZF, if X is infinite and $SC_r(X)$ holds, then X is Dedekind-infinite. More precisely, from a witnessing map f and any ordered $(N(r) + 1)$ -tuple of distinct elements of X , one can define an injection $\mathbb{N} \rightarrow X$.*

Proof. Let $S_0 \subseteq X$ have $N(r) + 1$ elements, equipped with the order supplied by the chosen tuple. Recursively put

$$S_{k+1} = H_f(S_k) = S_k \cup f[\mathcal{P}(S_k)].$$

Each S_k is finite. Since $|S_k| > N(r)$, [equation \(8.1\)](#) gives $|S_{k+1}| > |S_k|$.

It remains to choose a new point at every stage without invoking countable choice. We define an order on S_{k+1} canonically from the order on S_k . The latter induces a lexicographic order on the finite set $\mathcal{P}(S_k)$. For $y \in f[\mathcal{P}(S_k)]$, let $\mu_k(y)$ be the least subset $A \subseteq S_k$ with $f(A) = y$. The map μ_k is injective, so it transports the lexicographic order to a linear order on $f[\mathcal{P}(S_k)]$. Order S_{k+1} by listing the old points in their old order, followed by the new points in this transported order.

Let x_k be the first point of $S_{k+1} \setminus S_k$ in that order. The construction is recursive and functional, so Replacement produces the sequence $k \mapsto x_k$. The points are pairwise distinct because $x_k \in S_{k+1} \setminus S_k$ and the S_k are increasing. Hence this sequence is an injection from $\mathbb{N}$ into X . $\square$

Corollary 8.4 (Dedekind-finite obstruction). *No infinite Dedekind-finite set, and in particular no amorphous set, satisfies $SC_r(X)$ for any finite r .*

Remark 8.5 (more than existential infinitude). The proof does not appeal to the principle that a countable union of nonempty finite sets has a choice function. It manufactures a canonical local order from least preimages at each stage. Relative to f and the finite seed, the countable subset is definable. This feature is useful for formalization and may be stronger than the bare Dedekind-infinitude conclusion in symmetric models.

[write] 5 October 2026. The last sentence is a speculation; it is stated with the source’s definability question as item [F15](#) of [Section 13.1](#).

9 Antichain transfer and the duplicability obstruction

The second infinite mechanism extracts a finite-to-one map from a large, explicit antichain.

Definition 9.1 (antichain code). *An antichain code from U into X is a map*

$$\Theta : \mathcal{P}(U) \longrightarrow \mathcal{P}(X)$$

whose range is an antichain under inclusion.

[write] Read with “injective”, 5 October 2026. As printed the definition allows a non-injective, even constant, Θ , and then [Lemma 9.2](#) and [Theorem 9.4](#) are false. The repair is to require Θ to be injective, equivalently $\Theta(A)$ and $\Theta(B)$ incomparable whenever $A \neq B$; the counterexample and the check of every later use are in [Remark 9.10](#).

Lemma 9.2 (width becomes bounded multiplicity). *If $f : \mathcal{P}(X) \rightarrow X$ has fibers of width at most r and Θ is an antichain code from U into X , then $f \circ \Theta : \mathcal{P}(U) \rightarrow X$ is at most r -to-one.*

Proof. If $r + 1$ distinct subsets of U had the same image, their Θ -images would be $r + 1$ pairwise incomparable members of one fiber of f . $\square$

[write] 5 October 2026. The proof needs the $r + 1$ images to be distinct, that is, Θ injective; distinct sets may share one image under a code in the printed sense (Remark 9.10).

We use Forster’s theorem in its published ZF form.

Theorem 9.3 (Forster). *If there is a finite-to-one map $\mathcal{P}(U) \rightarrow U$, then U is finite.*

Forster’s result strictly strengthens Cantor’s theorem in the absence of Choice [3].

[write] Forster’s theorem, as read 5 October 2026. The write read the abstract and bibliographic data of [3] on the publisher’s page (Project Euclid; *J. Symbolic Logic* 68(4), December 2003, 1251–1253), not the body. The abstract reads “It is shown in ZF (without choice) that if there is a finite-to-one map $\mathcal{P}(X) \rightarrow X$, then X is finite”, with a surjection arrow. Theorem 9.3 is printed for an arbitrary map; it follows in ZF from the form for surjections. Given a finite-to-one $g : \mathcal{P}(U) \rightarrow U$, put $g'(\{u\}) = u$ for $u \in U$ and $g'(A) = g(A)$ for every A that is not a singleton. Then g' is onto, and $g'^{-1}(u) \subseteq \{\{u\}\} \cup g^{-1}(u)$ is finite, so g' is a finite-to-one surjection and U is finite. (The converse is trivial.) Both statements are theorems of ZF; “strictly strengthens” is read as: its hypothesis is weaker than Cantor’s, since an injection is finite-to-one, while without Choice a finite-to-one map need not yield an injection. Whether Forster’s paper says so was not checked.

Theorem 9.4 (antichain-transfer obstruction). *Let U be infinite. Suppose there are*

1. *an antichain code $\Theta : \mathcal{P}(U) \rightarrow \mathcal{P}(X)$, and*
2. *a finite-to-one map $p : X \rightarrow U$.*

Then $SC_r(X)$ fails for every finite r .

Proof. If f witnessed $SC_r(X)$, then $h = f \circ \Theta$ would be at most r -to-one by lemma 9.2. For a point $u \in U$, the fiber of $p \circ h$ is the union, over the finite set $p^{-1}(u)$, of fibers of h of size at most r . Hence $p \circ h : \mathcal{P}(U) \rightarrow U$ is finite-to-one, contradicting theorem 9.3. $\square$

[write] 5 October 2026. As printed, this theorem is false: $X = \{0, 1, 2\}$, $U = \mathbb{N}$, p the inclusion and $\Theta \equiv \emptyset$ satisfy its hypotheses, while $SC_r(X)$ holds for every $r \geq 1$ (Remark 9.10). With Θ injective the proof above is complete.

The required antichain code has a simple two-channel construction.

Lemma 9.5 (complementary-transversal code). *If there is an injection $e : U \times 2 \rightarrow X$, then there is an antichain code $\Theta : \mathcal{P}(U) \rightarrow \mathcal{P}(X)$.*

Proof. Write $e_i(u) = e(u, i)$. The two ranges are disjoint. Define

$$\Theta(A) = e_0[A] \cup e_1[U \setminus A].$$

If $A \neq B$, choose u in their symmetric difference. If $u \in A \setminus B$, then $e_0(u) \in \Theta(A) \setminus \Theta(B)$ while $e_1(u) \in \Theta(B) \setminus \Theta(A)$. Thus neither coded set contains the other. The other case is symmetric. $\square$

Theorem 9.6 (two-channel core criterion). *Let U be infinite. If*

$$U \times 2 \preceq X \quad \text{and} \quad X \preceq_{\text{fto}} U,$$

then $\text{SC}_r(X)$ fails for every finite r .

Proof. The first map supplies the antichain code by [lemma 9.5](#); the second map and [theorem 9.4](#) finish the proof. $\square$

Corollary 9.7 (binary duplicability obstruction). *If X is infinite and $X \times 2 \preceq X$, then $\text{SC}_r(X)$ fails for every finite r .*

Proof. Use [theorem 9.6](#) with $U = X$ and $p = \text{id}_X$. $\square$

Corollary 9.8 (well-orderable and choice cases). *No infinite well-orderable set satisfies $\text{SC}_r(X)$. Consequently ZFC proves that $\text{SC}_r(X)$ holds exactly for the finite nonempty sets satisfying $W_{|X|} \leq r|X|$.*

Proof. Every infinite well-orderable set is equipotent with an infinite initial ordinal κ , and $2 \times \kappa$ injects into κ . Apply [corollary 9.7](#). Under AC, every set is well-orderable. The finite cases are exactly [theorem 5.1](#). $\square$

Remark 9.9 (why Forster is the right dependency). Cantor’s theorem alone excludes an injection $\mathcal{P}(U) \rightarrow U$, but $f \circ \Theta$ is only r -to-one. Turning that map into an injection would silently use a finite choice operation on its fibers. Forster’s theorem is precisely the choice-free statement needed here.

Remark 9.10 ([write] the antichain code must be injective). Added 5 October 2026 (batch 114). As printed, Definition 9.1 asks only that the range of Θ be an antichain. With that definition Lemma 9.2 and Theorem 9.4 are false, and so is the sentence of the abstract that the antichain-transfer theorem “converts any antichain code $\mathcal{P}(U) \rightarrow \mathcal{P}(X)$ together with a finite-to-one map $X \rightarrow U$ into a finite-to-one map $\mathcal{P}(U) \rightarrow U$ ”. The printed statements are kept as delivered; they are to be read with the repaired definition below.

Counterexample. Let $X = \{0, 1, 2\}$, $r = 1$, $U = \mathbb{N}$, let $p : X \rightarrow \mathbb{N}$ be the inclusion, and let $\Theta(A) = \emptyset$ for every $A \subseteq \mathbb{N}$.

(i) *The hypotheses of Theorem 9.4 hold.* U is infinite. The range of Θ is $\{\emptyset\}$, a family with one member, hence an antichain; so Θ is an antichain code in the printed sense. The map p is injective, so each $p^{-1}(u)$ has at most one element and p is finite-to-one.

(ii) $\text{SC}_1(X)$ holds. The chains

$$C_0 : \emptyset \subset \{0\} \subset \{0, 1\} \subset \{0, 1, 2\}, \quad C_1 : \{1\} \subset \{1, 2\}, \quad C_2 : \{2\} \subset \{0, 2\}$$

have $4 + 2 + 2 = 8$ members, pairwise distinct, so they partition $\mathcal{P}(X)$. Put $f(A) = i$ for $A \in C_i$. Every fiber $f^{-1}(i) = C_i$ is a chain, of width 1. (This is Theorem 5.1 with $n = 3$, $W_3 = 3 \leq 1 \cdot 3$, and Theorem 7.1.) Since a width-1 fiber has width at most r , $\text{SC}_r(X)$ holds for every $r \geq 1$.

(iii) *The conclusion fails.* Theorem 9.4 asserts that $\text{SC}_r(X)$ fails for every finite r ; by (ii) it fails for none.

- (iv) *Where the proof breaks.* $f \circ \Theta$ is the constant map $\mathcal{P}(\mathbb{N}) \rightarrow X$ with value $f(\emptyset) = 0$, so its fiber over 0 is all of $\mathcal{P}(\mathbb{N})$; it is not at most 1-to-one, and $p \circ f \circ \Theta$ is not finite-to-one. The proof of Lemma 9.2 infers from “ $r + 1$ distinct subsets” that their images are “ $r + 1$ pairwise incomparable members”, which needs the images to be distinct. The lemma fails for every r : with the same f and $\Theta \equiv \emptyset$ on $\mathcal{P}(U)$ for any set U with more than r subsets, $f \circ \Theta$ is constant on more than r points.

Repair. Call $\Theta : \mathcal{P}(U) \rightarrow \mathcal{P}(X)$ an (injective) *antichain code* if Θ is injective and its range is an antichain; equivalently, $\Theta(A)$ and $\Theta(B)$ are incomparable whenever $A \neq B$. (If Θ is injective with antichain range, then $A \neq B$ gives two distinct members of the range, which are incomparable. Conversely, incomparable sets are distinct, so Θ is injective, and two distinct members of the range are $\Theta(A), \Theta(B)$ with $A \neq B$, hence incomparable.) With this definition:

- *Lemma 9.2 holds.* If distinct $A_0, \dots, A_r \subseteq U$ had $f(\Theta(A_i)) = x$ for all i , the sets $\Theta(A_0), \dots, \Theta(A_r)$ would be pairwise incomparable, hence pairwise distinct, members of $f^{-1}(x)$: an antichain of size $r + 1$ in a fiber of width at most r . Injectivity is used here, and only here.
- *Theorem 9.4 holds,* with its printed proof: for $u \in U$, $(p \circ h)^{-1}(u) = \bigcup_{x \in p^{-1}(u)} h^{-1}(x)$ is a union of finitely many sets with at most r elements, hence finite (in ZF, a finite union of finite sets is finite), and Theorem 9.3 applies.
- *Lemma 9.5* already proves that $\Theta(A)$ and $\Theta(B)$ are incomparable for $A \neq B$, so its code is injective and satisfies the repaired definition; nothing in it changes.
- *Theorem 9.6, Corollaries 9.7 and 9.8, Theorem 10.1.* Each uses Theorem 9.4 only with the code of Lemma 9.5 (Theorem 9.6 directly, the others through it), so all stand, and with them the complete ZFC classification and the counterexample profile.
- *Elsewhere.* The remark above (“ $f \circ \Theta$ is only r -to-one”) is true for injective codes. The dependency chain of Stage IV in Section 12 (“antichain code $\Rightarrow$ boundedly finite-to-one”) and the research items 12 and 16 of Section 13 must be read with injective codes. The abstract’s sentence quoted above and the row “Antichain-transfer theorem: Proved here” of the ledger in Appendix B are correct with the repaired definition only; the conclusion uses only the complementary-transversal code.

The repair loses nothing: the only antichain code the manuscript constructs is the injective one of Lemma 9.5, which is also the form in which the code appears in the MathOverflow thread (note after Theorem 7.1).

10 The remaining choiceless frontier

Combining the preceding sections yields a sharp profile of any counterexample to [conjecture 2.5](#).

Theorem 10.1 (counterexample profile). *Fix $r \geq 2$. If an infinite set X satisfies $\text{SC}_r(X)$, then:*

1. X is Dedekind-infinite, and a countable subset is definable from the witness and finitely many parameters;
2. $X \times 2 \not\preceq X$;
3. more generally, there is no infinite U with $U \times 2 \preceq X$ and $X \preceq_{\text{fto}} U$;

4. every finite $S \subseteq X$ of size n emits at least $q_r(n) - n$ labels outside itself whenever this number is positive;
5. the ambient model violates Choice.

Proof. Items (1)–(4) are [theorems 8.1, 8.3 and 9.6](#). Item (5) follows from [corollary 9.8](#). $\square$

This profile rules out the most familiar choiceless candidates. An amorphous set is impossible by (1); an infinite cardinal is impossible by (2); and a set that is finite-to-one reducible to an infinite two-channel core is impossible by (3). The only remaining region consists of Dedekind-infinite sets with highly nonclassical finite-multiple arithmetic.

10.1 Why the obvious reductions stop here

One might try to decompose each width- r fiber into r chains and apply the $r = 1$ theorem. There are two obstructions.

1. Infinite Dilworth decompositions can require compactness or choice.
2. Even if each fiber separately admits a decomposition, choosing one for every $x \in X$ may be a family-choice principle; afterward one still has an $X \times r$ -indexed family, and $X \times r$ need not reduce to X in ZF.

Likewise, restricting f to one rank produces an r -to-one map from a finite-subset functor into X , but boundedly finite-to-one maps do not in general canonically split into injections without Choice [7].

10.2 Two credible routes forward

A proof of [conjecture 2.5](#) could seek a *canonical* chain splitting special to families of subsets, avoiding arbitrary finite choices. A counterexample would likely require a permutation or symmetric model with a Dedekind-infinite, non-duplicable set X and an equivariant map on its full power set. The local expansion theorem imposes a severe support condition: large finite subcubes must acquire exponentially many distinct output atoms. Either route is concrete enough for a focused research program.

[write] 5 October 2026. Both routes are heuristics (“could seek”, “would likely require”); they are recorded with items [F1](#), [F2](#) and [F4](#) of Section [13.1](#).

11 Algorithms, exact checks, and certificate complexity

The accompanying script `verification.py` uses only Python’s standard library. Its role is diagnostic, not foundational: the proofs above are symbolic.

11.1 What was checked

The executed verifier performs the following exact checks.

1. It constructs the recursive symmetric-chain decomposition for $0 \leq n \leq 14$ and verifies coverage, disjointness, saturation, symmetry, and the chain count W_n .
2. For $1 \leq n \leq 10$ and $1 \leq r \leq 5$, it groups chains and verifies that the number of groups is $\lceil W_n/r \rceil$.
3. It checks exact rational monotonicity of W_n/n and monotonicity of $\delta(n)$ for $1 \leq n \leq 300$.
4. It generates the defect and threshold tables in this manuscript and a numerical diagnostic for [theorem 6.2](#).

All assertions passed. The full textual output is included as `verification_report.txt`.

[write] The program in this directory, 5 October 2026. The program is shipped as `code/verification.py` and its output as `data/verification_report.txt`. It writes `verification_report.txt` *beside itself*, in the directory of the script, so run in place it would create `code/verification_report.txt`; the README gives a rerun on a copy. Rerun so on 5 October 2026 it finished in under a second with all assertions passing and an output identical to the shipped one up to line endings. It checks finite instances only (items 1–3 above); item 4 prints the tables and evaluates the asymptotic formula numerically, which proves nothing about Theorem [6.2](#). Nothing in it concerns Sections [7–10](#).

11.2 Complexity

An explicit decomposition necessarily names all 2^n subsets, so its output size is exponential. The verifier runs in time polynomial in that output size. The lower certificate is much smaller only in representation, not in cardinality: it contains $W_n \asymp 2^n / \sqrt{n}$ middle-layer subsets.

For formal work, one can avoid materializing the entire certificate by proving the recursive SCD algorithm correct once and evaluating only the closed formula for the number of chains. This separates an executable instance certificate from a reusable theorem.

11.3 Pseudocode

```
SCD(0) := [[empty set]]
for new = 1, ..., n:
    output := []
    for chain C = [A_k, ..., A_t] in SCD(new-1):
        append [A_k, ..., A_t, A_t union {new}] to output
        if C has at least two members:
            append [A_k union {new}, ..., A_{t-1} union {new}] to output
    chains := output
return chains
```

The two output chains partition the two copies of the input chain in $\mathbb{B}_{n-1} \times \{0, 1\}$, which makes the inductive coverage proof especially short.

12 A ProveIt formalization blueprint

The repository snapshot inspected for this report distinguishes formal proofs, executable certificates, and unformalized research reports [\[8\]](#). The present project can follow that boundary explicitly.

[write] **A proposal, 5 October 2026.** Nothing of this section exists in the repository. The repository’s Lean workspace pins mathlib v4.32.0, whose Mathlib/Combinatorics/SetFamily/LYM.lean proves the LYM inequality and Sperner’s theorem (`IsAntichain.sperner`); no file of that mathlib mentions Dilworth’s theorem (search of the source tree), so the option of Stage I to “import a finite Dilworth theorem from mathlib” is not available at that version. Stage IV must use injective antichain codes (Remark 9.10).

12.1 Stage I: finite posets

A small generic library should define finite antichains, chain covers, width, and budgeted colorings. The principal theorem can be stated schematically as

```
theorem exists_width_budget_coloring_iff
  (P : FinitePoset) (budget : Fin q -> Nat) :
  (exists c, forall i, width (fiber c i) <= budget i) <=>
  width P <= sum budget
```

The upper implication can either import a finite Dilworth theorem from mathlib or prove the needed chain-partition form through bipartite matching. The latter produces a reusable algorithmic certificate.

12.2 Stage II: Boolean lattices

Formalize the recursive SCD directly. Its invariants are:

1. every chain is saturated;
2. bottom rank plus top rank is n ;
3. chains are pairwise disjoint;
4. their union is the entire Boolean lattice.

The chain count follows by intersecting with the middle layer. Sperner’s lower bound can be imported if available or proved through the LYM inequality; for this particular theorem the standard SCD itself also supplies an upper chain cover while the middle layer supplies the lower antichain.

12.3 Stage III: set-theoretic local expansion

In the first-order ZF development, define the assertion that a function’s fibers contain no $(r + 1)$ -element antichain. The restriction to a finite subcube should then instantiate the finite theorem. The hull recursion in [theorem 8.3](#) requires only finite powersets, least elements of finite linear orders, and ordinary recursion on ω .

12.4 Stage IV: antichain transfer

The complementary-transversal code is elementary and should formalize independently of cardinal arithmetic. The dependency graph is:

$$\boxed{U \times 2 \hookrightarrow X} \implies \boxed{\mathcal{P}(U) \rightarrow \mathcal{P}(X) \text{ antichain code}} \implies \boxed{\mathcal{P}(U) \rightarrow X \text{ boundedly finite-to-one}}$$

followed by a finite-to-one map $X \rightarrow U$ and Forster's theorem. This makes Forster's theorem a named interface rather than a hidden metatheoretic step.

12.5 Stage V: independent verification

A Rocq/Coq port of the finite certificate checker would be modest and useful. The set-theoretic theorem can remain Lean-first until a common encoding of finite-to-one cardinal comparison is available. The article should continue to mark all unformalized infinite dependencies explicitly.

13 Further research questions and topics

1. **The main ZF classification.** Prove or refute [conjecture 2.5](#) for one fixed $r \geq 2$, beginning with $r = 2$.
2. **Permutation-model construction.** Can a Fraenkel–Mostowski or symmetric-extension model contain a Dedekind-infinite, non-duplicable X with $\text{SC}_2(X)$? The map must satisfy the finite expansion bounds equivariantly.
3. **Weak choice strength.** Determine the weakest standard choice principle implying the full finite classification for every r . Candidates include finite-choice schemes, the Boolean prime ideal theorem, and compactness principles sufficient for finite-width chain decompositions.
4. **Canonical chain splitting.** Does every width- r subfamily of a power set admit, in ZF, a decomposition into r chains after adjoining a finite amount of structure canonically definable from the family?
5. **Finite-to-one core extraction.** Does $\text{SC}_r(X)$ itself imply the existence of an infinite U with $U \times 2 \preceq X$ and $X \preceq_{\text{fto}} U$? A positive answer would settle the main conjecture through [theorem 9.6](#).
6. **Matching formulation.** The values $f(\{x, c_n\})$ from a definable countable core generate a bipartite graph with infinite left degree and bounded right degree. Identify the exact choice principle needed for a two-matching saturating the left side.
7. **Variable fiber budgets.** For an infinite target, replace the constant r by a function $b : X \rightarrow \mathbb{N}$. The finite restriction gives $W_n \leq \sum_{x \in f[\mathcal{P}(S)]} b(x)$. Which growth or summability hypotheses force nonexistence in ZF?
8. **Stability.** Classify finite colorings of $\mathbb{B}_n$ using $\lceil W_n/r \rceil + o(W_n)$ labels. Must most middle-layer classes be nearly saturated, and must most fibers be close to unions of r chains?
9. **Enumeration.** Count optimal labeled and unlabeled r -width compressions of $\mathbb{B}_n$. Even the subclass arising from grouped symmetric-chain decompositions has a nontrivial orbit structure.
10. **Other Peck posets.** For any finite poset the numerical optimum is width divided by r , but Boolean lattices offer canonical certificates. Develop equally explicit constructions for products of chains, subspace lattices, Young lattices at fixed rank, and divisor lattices.
11. **q -ary cubes.** Replace $\mathcal{P}([n])$ by $[q]^n$ under coordinate order. Determine exact self-indexed thresholds when the target has size n , qn , or the size of a prescribed rank.

12. **Transfinite width bounds.** Let the allowed fiber width be an infinite cardinal ρ . Under which assumptions does an antichain code convert a ρ -width map into a cardinal inequality strong enough to contradict generalized Cantor theorems?
13. **Reverse mathematics of the finite proof.** Over weak arithmetic, calibrate the strength of the uniform statement that every finite poset obeys [theorem 3.1](#), distinguishing existence from an executable chain decomposition.
14. **Formalize Forster.** Produce a self-contained Lean proof of the theorem that a finite-to-one $\mathcal{P}(X) \rightarrow X$ forces X finite, with an explicit audit of every finite-choice step avoided by the argument.
15. **Definability strength.** Analyze the sequence produced in [theorem 8.3](#): can the finite seed be removed, or can one obtain a parameter-free countable subset from structural assumptions on f ?
16. **Benchmark variants.** Package the finite theorem, the threshold inversion, and the antichain-transfer lemma as separate automatically verifiable challenge problems. A high-end version can ask solvers to discover the Dedekind-infinitude recursion without being told the hull operator.

13.1 Further questions and research

[write] **Added 5 October 2026, batch 114.** Vladimir’s standing rule of 4 October 2026: every claim of the source that is not proved in the text is stated here as an open question, with its source, the sketch it came with, and what is missing. The one demonstrably wrong statement is corrected in Remark [9.10](#), not here. The items are Conjecture [2.5](#) (item [F1](#)), the fifteen other research items of the list above, and one item on the remainder of Theorem [6.2](#). Proposition [13.1](#) below proves the graph-theoretic assertion of item [6](#).

- F1. *The ZF classification for $r \geq 2$* (Conjecture [2.5](#); item [1](#)). Proved: the finite case in ZF (Theorem [5.1](#), for every r), the whole statement in ZFC (Corollary [9.8](#)), and $r = 1$ in ZF (Glazer, imported, Theorem [7.1](#)). Missing: an infinite X with $\text{SC}_r(X)$, $r \geq 2$, in a model of ZF, or a proof that none exists. Theorem [10.1](#) says what such an X must look like: Dedekind-infinite, $X \times 2 \not\leq X$, no infinite U with $U \times 2 \leq X$ and $X \preceq_{\text{fto}} U$, and Choice fails. A positive answer to item [F5](#) would settle it; the two routes of Section [10.2](#) are heuristics.
- F2. *A permutation or symmetric model* (item [2](#)). The source’s sketch: the map must be equivariant and must satisfy the expansion bounds of Theorem [8.1](#); its sentence that a counterexample “would likely require” such a model is a heuristic. No model is proposed.
- F3. *The weakest choice principle* (item [3](#)). The candidates named are finite-choice schemes, the Boolean prime ideal theorem and compactness principles. Note (write): by Corollary [9.7](#) and Theorem [5.1](#), the principle “ $X \times 2 \leq X$ for every infinite X ” already yields the full classification for every r , and by Theorem [9.6](#) so does “every infinite X has an infinite U with $U \times 2 \leq X$ and $X \preceq_{\text{fto}} U$ ”. How these sit among the named candidates was not examined (literature not consulted).
- F4. *Canonical chain splitting* (item [4](#)). Missing: any construction; Section [10.1](#) explains why the obvious one (Dilworth fiber by fiber, then $r = 1$) needs choice.
- F5. *Finite-to-one core extraction* (item [5](#)). Does $\text{SC}_r(X)$ give an infinite U with $U \times 2 \leq X$ and $X \preceq_{\text{fto}} U$? A yes settles item [F1](#) through Theorem [9.6](#). Theorem [8.3](#) gives a definable countable

subset; what is missing is a finite-to-one map from all of X onto a set that carries two disjoint copies of itself.

- F6. *Matching formulation* (item 6). The source asserts without proof that the values $f(\{x, c_n\})$ give a bipartite graph with infinite left degree and bounded right degree; Proposition 13.1 below proves this, with right degree at most $2r$. Open: the exact choice principle needed for a “two-matching saturating the left side”; the source does not define the term, and the write did not fix a definition.
- F7. *Variable fiber budgets* (item 7). The source’s finite restriction $W_n \leq \sum_{x \in f[\mathcal{P}(S)]} b(x)$ is the necessity half of Theorem 3.1 applied to f restricted to $\mathcal{P}(S)$, with the labels in $f[\mathcal{P}(S)]$ as bins and capacities $b(x)$ (an antichain of size W_n meets the fiber of x in at most $b(x)$ points). Open: which growth or summability conditions on b force nonexistence in ZF.
- F8. *Stability* (item 8); no sketch.
- F9. *Enumeration* (item 9); the sentence that grouped symmetric-chain decompositions have “a nontrivial orbit structure” is not substantiated.
- F10. *Other Peck posets* (item 10). The numerical optimum $\lceil \text{wd}(P)/r \rceil$ is Corollary 3.2 for every finite poset; open are explicit certificates of the kind of Section 4.
- F11. *q -ary cubes* (item 11): the numbers follow from Corollary 3.2 once the width of $[q]^n$ is known; the exact self-indexed thresholds are not computed.
- F12. *Transfinite width bounds* (item 12), with injective codes (Remark 9.10); no sketch.
- F13. *Reverse mathematics of Theorem 3.1* (item 13); no sketch.
- F14. *Formalize Forster* (item 14); a proposal. Neither Forster’s proof (not read by the write) nor any formalization of it is in the repository.
- F15. *Definability strength* (item 15), together with the speculation of Remark 8.5 that the definable countable subset “may be stronger than the bare Dedekind-infinitude conclusion in symmetric models”. No model or argument is given.
- F16. *Benchmark variants* (item 16); a proposal. An antichain-transfer challenge must state the injective form (Remark 9.10).
- F17. *An explicit remainder in Theorem 6.2* (write). The source’s proof ends with one sentence (“a bounded rounding and parity term”) and gives no constant. Missing: explicit bounds for $N(r) - L - \frac{3}{2} \log_2 L - \frac{1}{2} \log_2 \frac{\pi}{2}$; the values observed for $r = 2^k, 4 \leq k \leq 60$, lie between -0.5636 and 1.6743 (note after the theorem).

Proposition 13.1 ([write] the graph of item 6). *Let $r \geq 1$, let $f : \mathcal{P}(X) \rightarrow X$ witness $\text{SC}_r(X)$, and let $c : \mathbb{N} \rightarrow X$ be injective (for instance the sequence of Theorem 8.3). Let G be the bipartite graph whose two sides are copies of X , with x on the left joined to y on the right if and only if $y = f(\{x, c_n\})$ for some n with $c_n \neq x$. Then every left vertex has infinitely many neighbours and every right vertex has at most $2r$ neighbours. G is definable from f and c .*

Proof. Fix $x \in X$. Since c is injective, $c_n = x$ for at most one n , so $N_x = \{n : c_n \neq x\}$ is infinite, and $n \mapsto \{x, c_n\}$ is injective on N_x . The sets $\{x, c_n\}$ ($n \in N_x$) are distinct two-element sets, hence pairwise incomparable, so each fiber of f contains at most r of them. If x had only finitely many

neighbours $y_1, \dots, y_m$, all of these infinitely many sets would lie in $f^{-1}(y_1) \cup \dots \cup f^{-1}(y_m)$, which contains at most mr of them; a contradiction.

Fix $y \in X$. The two-element sets in $f^{-1}(y)$ are pairwise incomparable, so there are at most r of them, and their union T_y has at most $2r$ elements. If x is joined to y , then $\{x, c_n\} \in f^{-1}(y)$ for some n , so $x \in T_y$. Hence y has at most $2r$ neighbours. The definition of G uses only f and c . $\square$

14 Conclusion

Bounded-width power-set compression is a compact problem with three distinct layers. The finite layer is exact and universal: width is the only resource, and a middle antichain plus a grouped chain decomposition gives a complete certificate. For Boolean lattices this produces the central-binomial threshold and its logarithmic inversion. The classical $r = 1$ case is not merely a finite observation; Glazer’s theorem shows that the same cutoff is absolute in ZF.

For larger r , the infinite problem exposes precisely where choice enters. The local finite theorem still acts inside every finite subcube and forces a canonical countable subset. A large complementary-transversal antichain, combined with Forster’s theorem, eliminates every binary-duplicable set and therefore every well-orderable infinite set. What remains is not a vague “pathological set” possibility but a sharply delimited class of Dedekind-infinite, non-duplicable objects with unusual finite-multiple arithmetic.

This combination – exact finite extremal structure, an objective certificate surface, a clean ZFC classification, and a narrow unresolved choiceless boundary – makes the problem well suited both to Glazer’s research themes and to ProveIt’s formal research program.

A Exact tables

Table 1: Central width and least self-indexed defect.

n	$W_n = \binom{n}{\lfloor n/2 \rfloor}$	$\delta(n) = \lceil W_n/n \rceil$
1	1	1
2	2	1
3	3	1
4	6	2
5	10	2
6	20	4
7	35	5
8	70	9
9	126	14
10	252	26
11	462	42
12	924	77
13	1,716	132
14	3,432	246
15	6,435	429
16	12,870	805
17	24,310	1,430
18	48,620	2,702
19	92,378	4,862
20	184,756	9,238
21	352,716	16,796
22	705,432	32,066
23	1,352,078	58,786
24	2,704,156	112,674

Table 2: Selected exact thresholds $N(r)$.

r	$N(r)$	r	$N(r)$
1	3	26	10
2	5	32	10
3	5	64	11
4	6	128	12
5	7	256	14
8	7	512	15
9	8	1,024	16
16	9	4,096	18
65,536	23	1,048,576	27

B Claim and dependency ledger

Claim	Status	Dependency or verification
Budgeted finite-poset theorem	Proved here	Dilworth’s finite chain-decomposition theorem.
Exact Boolean formula	Proved here	Sperner’s width theorem plus the budget theorem; SCD gives explicit upper certificate.
Finite self-indexed classification	Proved here	Immediate from the exact Boolean formula.
Monotonicity and asymptotics	Proved here	Exact binomial ratios and Stirling’s formula.
Glazer chain-fiber classification	Imported	MathOverflow answer of 2 December 2025; not reproved in full.
Exact local expansion	Proved here	Restriction to a finite Boolean subcube.
Definable	Proved here	Iterated finite hull with canonical least-preimage orders.
Dedekind-infinitude	Imported	JSL 68 (2003), 1251–1253.
Forster theorem	Imported	Width-to-multiplicity lemma plus Forster.
Antichain-transfer theorem	Proved here	Explicit complementary-transversal antichain code.
Binary duplicability obstruction	Proved here	Binary duplicability of infinite well-orderable sets plus finite theorem.
Full ZFC classification	Proved here	Binary duplicability of infinite well-orderable sets plus finite theorem.
Full ZF classification for $r \geq 2$	Open in this manuscript	Stated as conjecture 2.5 ; no solution claimed.
Historical novelty	Not established	Targeted source search only; independent literature review required.
Formal verification	Not claimed	Python checks finite instances; no Lean/Rocq proof accompanies the article.

[write] 5 October 2026. “Proved here” means proved in this text. The row “Antichain-transfer theorem: Proved here” holds only with injective antichain codes (Remark 9.10); the rows that depend on it (binary duplicability, full ZFC classification) stand, because the code they use is injective. Glazer’s answer is dated 2 December 2025 on MathOverflow, as the ledger says. The write adds Proposition 13.1 (proved in Section 13.1).

C Reproducing the checks

From the extracted archive, run

```
python3 verification.py
```

The script rewrites `verification_report.txt` and exits nonzero if any assertion fails. No third-party Python package is required. To rebuild the article, run

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

or use the PDF skill's `latex_to_pdf.py` wrapper.

[write] In this repository, 5 October 2026. The “PDF skill's `latex_to_pdf.py` wrapper” is a tool of the environment in which the manuscript was generated; it is not part of the delivery or of ProveIt, and the sentence is kept as delivered. Use `latexmk` as above (the shipped PDF was built so, with `pdfLATEX`). In the repository the program is `code/verification.py` and it rewrites `verification_report.txt` beside itself, so run it on a copy; the README gives the commands.

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